# TrailQuest

## Intelligent Hiking Companion

*Complete Full-Stack Web Application Proposal*

## 1. Executive Summary

TrailQuest is a full-stack hiking platform for Sri Lanka that combines hiking trails, equipment rentals, parking, offline maps, budget planning, weather forecasting, and community reviews into one application.

## 2. Problem Statement

Hikers currently depend on multiple disconnected apps. TrailQuest provides one intelligent platform for the complete hiking journey.

## 3. Objectives

- Provide hiking trail information
- Support tent & gear rentals
- Show nearby vehicle parking
- Offer offline trail maps
- Calculate hiking budgets
- Provide live weather
- Build a hiking community

## 4. Core Features

| Module | Description |
|---|---|
| Hiking Locations | Search trails by district, elevation and difficulty |
| Equipment Rental | Reserve tents, backpacks and camping gear |
| Parking | Locate secure parking with pricing |
| Offline Maps | Download GPX routes for offline navigation |
| Budget Planner | Estimate transport, food and rental costs |
| Weather | Rain, wind and temperature forecasts |
| Photos & Reviews | Community ratings and hiking memories |

## 5. Technology Stack

| Layer | Technology |
|---|---|
| Frontend | React + Vite + Tailwind CSS |
| Backend | Node.js + Express |
| Database | PostgreSQL |

| | |
|---|---|
| Maps | Leaflet + OpenStreetMap |
| Weather | Open-Meteo API |
| Authentication | JWT |

## 6. Database Tables

- Users
- Trails
- Rentals
- Parking
- Trips
- Expenses
- Reviews
- Photos

## 7. Development Timeline

| Week | Task |
|---|---|
| 1 | UI/UX + React |
| 2 | Authentication + Database |
| 3 | Trail Explorer + Maps |
| 4 | Budget + Weather |
| 5 | Rental Marketplace |
| 6 | Testing & Deployment |

## 8. Future AI Features

- AI trail recommendations
- Landslide & weather alerts
- Personalized hiking planner
- Plant recognition from photos

## 9. Expected Outcome

A scalable hiking platform that promotes eco-tourism while improving safety and convenience for hikers.